



Dear Search Committee Members,

I am writing to apply for the position of Assistant Professor of Population and Public Health Sciences at the Center for Genetic Epidemiology at USC. As a statistical geneticist, my work focuses on using population scale genomic datasets to understand the genetic basis of complex disease.

My work as a doctoral student at UCLA in the labs of Dr. Kirk Lohmueller and Dr. Sriram Sankararaman focused on developing new statistical methods to understand human genetics and evolution. With Dr. Lohmueller, I worked to understand why phenotype prediction models do not work equally well across human populations of diverse ancestries. With Dr. Sankararaman, I developed new methods to understand the role of archaic introgression in shaping the genomes of modern humans and found a substantial contribution of DNA from an unsampled archaic population. More recently, as a postdoctoral research fellow at Harvard Medical School, I have developed statistical tools to infer the genomic history of early modern humans in the lab of Dr. David Reich. In addition, I am co-advised by Dr. Alkes Price, where I am working on understanding the role gene-environment interactions play in complex phenotypes.

My research program at USC would continue in these directions and has the goal of leveraging genetic diversity to understand the genetic influences of complex disease. Students and postdocs in my group would be encouraged to work on developing quantitative methods to understand the genetic architecture of complex disease across diverse populations. I would be eager to collaborate both within and outside of the department and look forward to mentoring a diverse group of students and postdocs and fostering an inclusive community.

You may contact Dr. David Reich (reich@genetics.med.harvard.edu), Dr. Alkes Price (aprice@hsph.harvard.edu), and Dr. Nick Patterson (nickp@broadinstitute.org) to obtain letters supporting my application. The Center for Genetic Epidemiology would be an excellent environment to start my lab. I have attached a summary of my research interests to this cover letter. I am excited to submit this application and look forward to hearing from you. Thank you for your consideration.

Sincerely,
Arun Durvasula, PhD

Leveraging diverse ancestry and environmental variables to understand complex disease

Research Statement: Arun Durvasula

My main research focus is to leverage genetic diversity to understand the genetic influences of complex disease. Much work has focused on discovering disease associated genetic variation in populations of European descent. The extent to which these genetic variants are shared between populations remains poorly understood, especially for historically underrepresented populations. New datasets that focus on individuals of non-European ancestry open the door to answering this question and provide a crucial starting point for developing methods that model the genetic architecture of complex disease across diverse ancestries. My research program consists of three main directions. First, I will develop models that integrate population genetic insights to understand the global distribution of disease-causing genetic variation. Second, I will develop methods that integrate environmental information into disease risk prediction models. Third, I will develop methods that leverage whole genome sequencing data to move beyond single nucleotide variants to probe the effects of structural variation on complex disease. Developing statistical models that jointly consider the evolutionary history of populations around the world is crucial for ensuring equity in healthcare. My research program would build on my background in statistical and population genetics and extensive experience in analyzing genomic data. My contributions to the field so far have included:

- Understanding how population-specific genetic variation can impact the cross-population transferability of polygenic risk scores (**Durvasula** and Lohmueller 2021 *American Journal of Human Genetics*)
- Statistical modeling to understand deep human history and its impacts on modern human genetic variation (**Durvasula** and Sankararaman 2020 *Science Advances*)
- Developing a method for local ancestry inference from unknown source populations (**Durvasula** and Sankararaman 2019 *PLOS Genetics*).

1. Polygenic modeling of complex disease in diverse populations

A primary research goal of mine moving forward is to develop models of complex disease that improve risk prediction across populations. If achieved, this aim would improve patient care for individuals of diverse ancestries and address important equity issues in healthcare. Many studies have focused on the problem of cross population PRS accuracy across broad population labels such as “East-Asian” or “African”. However, these broad labels fail to fully describe the wide genetic variation observed in humans and the diverse demographic and population genetic processes that give rise to the genetic variation. For example, founder populations, which are those that have experienced extreme population bottlenecks, remain understudied from the perspective of polygenic risk.

My work in this area would seek to understand how genetic architecture changes under the different amounts of genetic drift and negative selection experienced by many populations around the world. Further, I would seek to develop new methods for fine-mapping causal variants in these populations, where the linkage disequilibrium landscape can be much different from European populations. Finally, I would work on modeling disease gene discovery in these populations, where genetic drift causes the causal allele frequency distribution to look quite different from European populations.

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2. Integrating information from biobank and electronic health record datasets to understand gene-environment interactions

Another research focus will be to understand the role the gene-environment interactions play in complex diseases. If successful, this would advance the goals of precision medicine by identifying specific combinations of genetic and environmental factors that are predictive of future disease risk. As a postdoctoral scholar, I have developed an approach to detect gene-environment interactions based on a regression of PRS multiplied by environmental variables against phenotypes. This approach has discovered >30 gene-environment interactions. Further, I found cases where the heritability of the complex disease or trait differs as a function of the environmental variable. This finding motivates future work in understanding the underlying causes of gene-environment interactions and developing statistical models that capture the spectrum of relationships between genetic variants and environmental variables.

A primary research aim moving forward is to improve PRS by explicitly modeling environmental variables in the PRS training procedure. A further aim is to develop methods for detecting gene-environment interactions in admixed populations. The PRS based approach I am currently taking does not account for differential accuracy between populations, which could lead to false positives because of ancestry co-varying with environmental variables. Finally, I would work to disentangle direct genetic effects from indirect genetic effects, which have been shown to affect the heritability of some traits like educational attainment.

3. Methods for genotyping and connecting structural variation to disease in diverse populations

Another focus area of my research program will be to develop tools to assay and understand the effects of structural variation on complex disease. If successful, this research would uncover a new set of genetic variants that cause disease within and across populations and potentially identify new pathways that are important for disease biology. Structural variation has long been hypothesized to have large phenotypic effects based on the larger physical impact on DNA compared to SNPs. However, challenges in assaying structural variation in large cohorts has limited our ability to test this hypothesis.

My plan for this research direction is to develop new methods for genotyping structural variants based on pan-genome graphs, which leverage a richer source of data for genotyping, and imputing structural variants based on coalescent methods, which leverage the shared genealogical history of SNPs and structural variants. These combined methods would be able to genotype structural variants in large panels of whole genome sequencing data (for example, UKBB=500K and AllofUS=100K) and impute them into cohorts where only genotype chip data is available (for example AllofUS=60K). With a large set of structural variants imputed, I will study patterns of association and understand how much heritability comes from common structural variants. In addition, studying the patterns of LD will allow for fine mapping of structural variants. Finally, it is possible to study the selective effects of structural variation by modeling the allele frequency distribution. Developing methods that focus on structural variation would allow us to contrast the range of selective effects across diverse types of genetic variation.